

Global Fractal Correlations in GW Noise: Methodological Reassessment of QW-1660 (v5–v76)

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2026-03-04

Abstract

This manuscript replaces the previous “empirical validation” wording with a strict methodological reassessment of the QW-1660 raw-strain program. We audit results from v5–v76 and later strict method checks (QW-1724, QW-1725, QW-2116), and separate robust findings from protocol-sensitive findings.

The central conclusion is conservative: **a stable global cross-detector fractal FIN correlation in raw LIGO/Virgo strain is not currently established at strict methodological level**. Several intermediate analyses reported nontrivial anti-persistent structure (e.g., cross-H values below 0.5), but later tests showed strong dependence on filtering, estimator choices, null construction, and protocol consistency. The current evidential status is **mixed/inconclusive**, with specific anomaly channels classified as non-robust in strict reanalysis.

1 Scope and Data

We analyze the already produced project artifacts, not new claims:

- QW-1660 series (v5–v76) JSON outputs,
- strict GW method audits (QW-1724, QW-1725, QW-2116),
- Phase 76 “true functional” run (whitened H1-L1, MF-DFA $q=0$),
- external blind GW holdout branch (QW-2078, QW-2077) reported separately from raw-strain global-correlation claims.

2 Protocol Inventory and Key Quantitative Points

Table 1 summarizes the critical numbers that drive the updated verdict.

Table 1: Key methodological checkpoints used in the reassessment

Artifact	What it reports	Key value(s)
QW-1660 v5	PSD scaling anomaly in selected setup	$Z = 6.29$
QW-1660 v13	Cross-detector coherence check	$r = -0.145$, verdict: no coherence
QW-1660 v24	Cross-Hurst vs shuffle/phase surrogates	H1-L1: 0.361 vs 0.531/0.397
QW-1660 v26	FIN multifractal summary	$H_0 = 0.2277$ (model-dependent stage)
QW-1660 v47	Filter paradox test	unfiltered 0.0373, filtered 0.1002
QW-1660 v57	Null MC (simplified model)	obs 0.311, “5 sigma” claim
QW-1660 v61 (strict rerun)	Full null with aligned protocol ($n = 131072$, 200 trials)	mean 0.3875, obs 0.4086, $p_{2s} = 0.398$
QW-1660 v63 (strict rerun)	Whitening test with null calibration ($n = 131072$, 200 trials)	whitened cross-H 0.7582, $p_{2s} = 0.00995$
Phase 76	Whitening test variant B (GWOSC fetch + MF-DFA $q=0$)	$H_{q=0} = 0.5051$
QW-1660 v65 (strict rerun)	Micro time-shift with permutation null ($n = 131072$, 200 perms)	best $ r = 0.0644$ at -24.66 ms, $p_{abs} = 0.00498$; not near 10 ± 2 ms
QW-1724	Method audit across v47–v65	verdict: high-risk inconclusive
QW-1725	Strict cross-H reanalysis	verdict: anomaly not robust
QW-2116	Method-repair gate over v61/v63/v65 strict branch	verdict: method repair pass; scientific anomaly verdict unchanged (non-robust)

3 Methodological Findings

3.1 What appears reproducible

- Anti-persistent behavior ($H < 0.5$) can be produced in multiple processing paths.

- Cross-detector statistics can depart from shuffle nulls in selected configurations.

3.2 What fails strict robustness

- **Protocol drift and internal inconsistency:** QW-1724 identifies critical contradictions (projection inconsistency, scale-invariance failure, hard-coded observation risk in one null test branch).
- **Method-repair versus scientific claim:** QW-2116 confirms protocol repair (no hard-coded observation, aligned n , calibrated null/permutation tests), but the scientific branch verdict remains `GW_CROSS_HURST_ANOMALY_NOT_ROBUST`.
- **Pair inconsistency:** strict reanalysis shows very different H1-L1 vs H1-V1 behavior under matched protocol, weakening “global” interpretation.
- **Lag inconsistency:** even when a local $\max|r|$ appears in strict rerun, it is off the expected light-travel window and therefore does not support a physically coherent broadband shared-phase signature.
- **Whitening dependence:** different whitening implementations produce substantially different outcomes (e.g., v63 vs Phase 76), indicating estimator/pipeline sensitivity.

3.3 Strict synthesis

Under strict criterion “same conclusion under controlled protocol variants and nulls,” the raw-strain global FIN-correlation claim is not closure-ready. Current best strict label is:

$$\text{GW_RAW_STRAIN_GLOBAL_FIN_STATUS} = \text{INCONCLUSIVE} / \text{NOT_ROBUST} \quad (1)$$

4 Relation to the External Blind GW Holdout Branch

A separate branch (QW-2078 $\rightarrow$ QW-2077) reports threshold pass on a locked holdout feature protocol:

$$\text{AUC} = 0.81496, \quad \text{ADV}^{q90} = 0.31028, \quad \text{SEP}_{\text{median}} = 0.002056, \quad \text{GAP}_{\text{median}} = 0.001289.$$

All preregistered thresholds pass in that branch.

This result is important, but it is **not identical** to proving a universal raw-strain global fractal background. It supports a specific locked prediction test, not full closure of the broader Phase-16 global-correlation claim.

5 Updated Scientific Verdict

1. The prior wording “empirical validation of global FIN background” was too strong.
2. The raw-strain QW-1660 line shows nontrivial effects but also major methodological fragility.

3. QW-2116 confirms that methodology was repaired, but it does not upgrade the scientific anomaly verdict.
4. Strict current status for global raw-strain FIN correlation is mixed/inconclusive.
5. Claims must remain falsifiable and should be phrased as provisional until independent replication under locked protocol is completed.

6 Minimum Protocol for a Future Decisive Test

- Pre-register one primary estimator and one fallback estimator (no post-hoc switching).
- Lock GPS/event list and detector-pair selection before running.
- Lock whitening/filter pipeline and edge-cropping rules.
- Use multiple null families (shuffle, phase-randomized, PSD-matched synthetic, circular shifts) with fixed trial count.
- Publish complete lag profiles and pairwise consistency statistics.
- Require agreement across independent environments and independent teams.

Reproducibility Pointers

- QW-1660 artifacts: `QW_1660_v*.json`
- Strict audits: `report_qw1724_gw_method_audit.json`, `report_qw1725_gw_strict_cross_hurst_reanalysis.json`, `report_qw2116_gw1660_method_repair_gate.json`
- Strict rerun scripts: `phase61_full_null_model_strict.py`, `phase63_whitening_test_strict.py`, `phase65_micro_timeshift_strict.py`
- Phase 76 artifacts: `FIN_Search_LIGO_Phase76_TrueFunctional.py`, `phase76_true_fin_results.json`